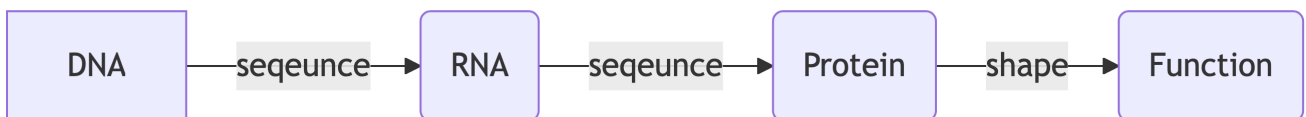


CSC 483 - Applied Biological Data Science W3D1

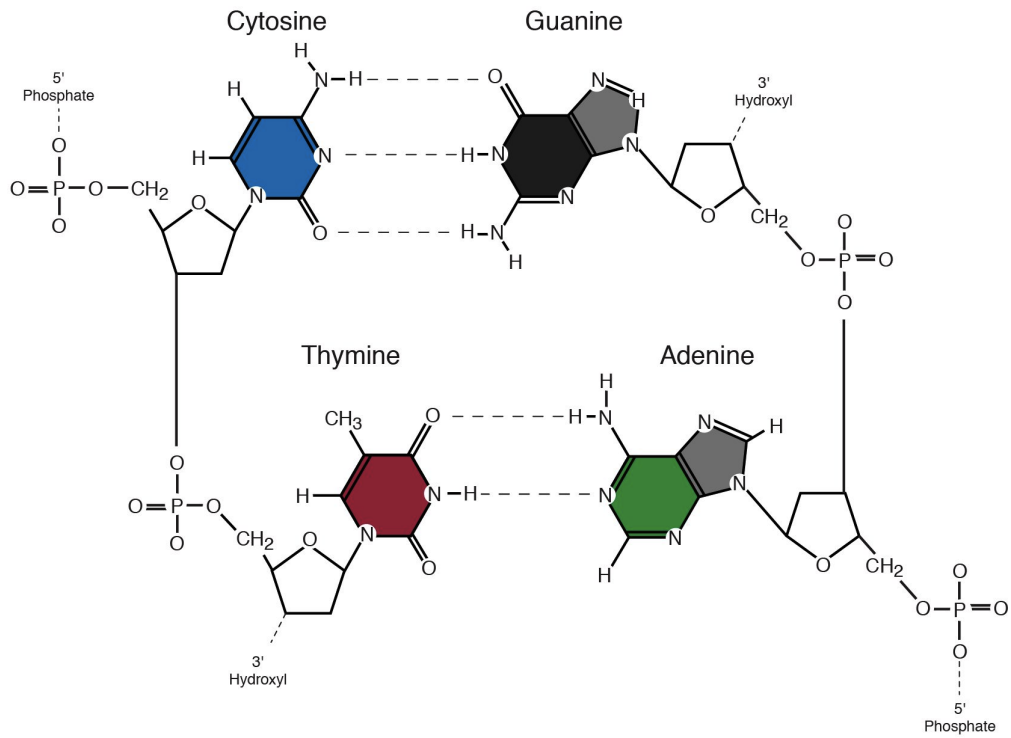
CSC 483: Applied Biological Data Science

- Warm up: (15')
- Sequencing data (30')
 - Gene sequence interpretation
 - Gene content vs. expression
 - Analysis goals: gene function, use
- Activity: metadata summaries (45')
- Present results, brainstorm ideas (15')
- HW:
 - rosalind problems on sequence analysis (w/ colab notebooks?)
 - pull request of e coli metadata analysis
 - due: tues

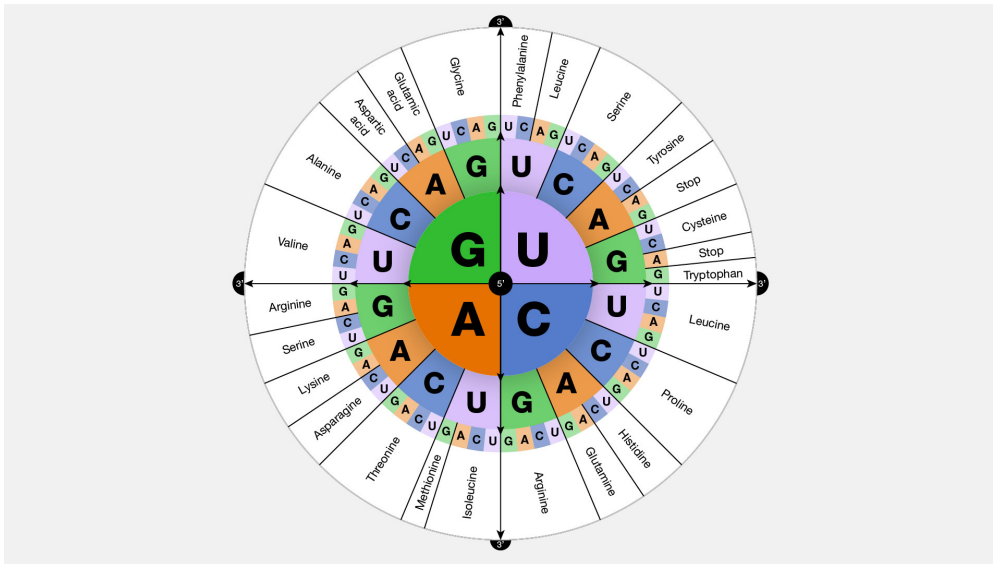
What is Biological Data?



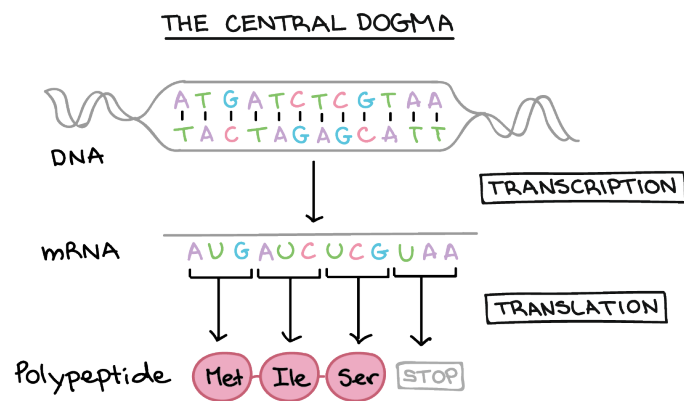
Rooted in Chemistry



Ends up being about information



What is a 'gene'



Biological Data Accumulation

Number of entries in biological sequence databases

Biological sequence databases store data such as DNA, RNA, and amino acid sequences and 3D protein structures. This data includes entries from GenBank, RefSeq, PDB, UniProtKB/SwissProt, as well as predicted protein structures in AlphaFoldDB and ESMAtlas.

Our World
in Data

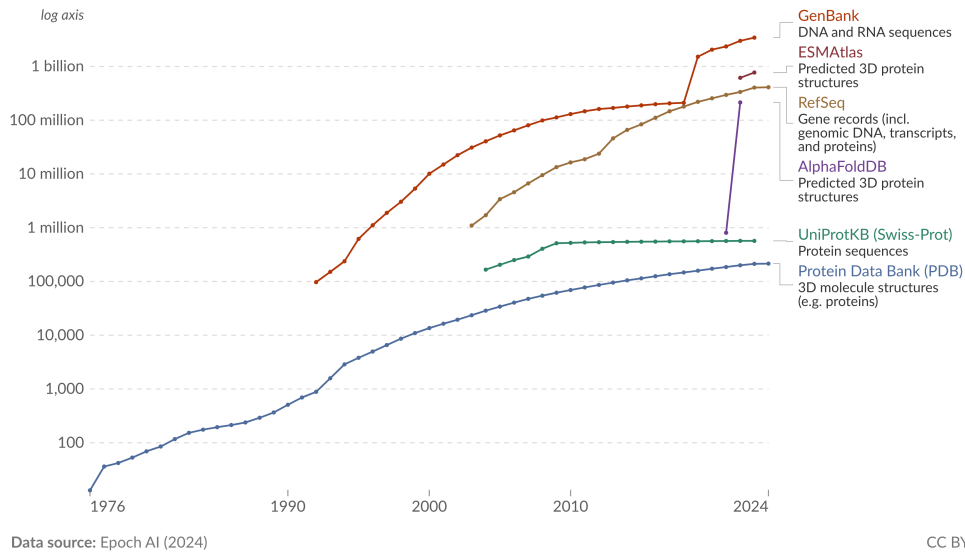


Figure 1: Biological Data

<https://ourworldindata.org/grapher/number-of-entries-in-biological-sequence-databases>

Ethics of Data

- measured vs predicted data?
- references?
- funding sources?
- reproducibility?
- data types?
- cross population?
- intention of the data post? intention of the original data source?

Purpose of Collecting Biological Data

- for human data, studying health and disease, e.g. 1000s Genomes Project: <https://www.internationalgenome.org/>
 - detect disease-associated alleles
 - check specific loci, spots
 - RNAseq - does it cause disease

IGSR: The International Genome Sample Resource
Supporting open human variation data

Home About Data Help Search IGSR

The International Genome Sample Resource

The 1000 Genomes Project created a catalogue of common human genetic variation, using openly consented samples from people who declared themselves to be healthy. The reference data resources generated by the project remain heavily used by the biomedical science community.

The International Genome Sample Resource (IGSR) maintains and shares the human genetic variation resources built by the 1000 Genomes Project. We also update the resources to the current reference assembly, add new data sets generated from the 1000 Genomes Project samples and add data from projects working with other openly consented samples.

Explore the data sets in IGSR through our data portal

Access HGSVC data

View variants in genomic context in Ensembl

Population	Size (N)	Genetic Diversity (H)	Genetic Diversity (S)	Genetic Diversity (I)
AFR	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
ACB	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
ASW	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
CEU	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
FIN	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
GBR	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
IBS	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
JPT	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
KAS	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
MDL	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
PEL	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
STP	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
TSI	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)
YRI	1,092	0.0001 (100)	0.0001 (100)	0.0001 (100)

Purpose of Collecting Biological Data

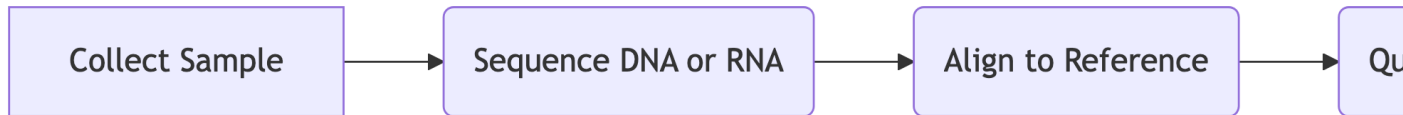
- for non-human data (microbes, animals): who is there and what are they doing?
 - whole genomes
 - 16S, ITS
 - meta-genomes
 - RNAseq - what are they doing
 - meta-transcriptomics

Slides on gene expression and the environment

- always need comparisons

- treat cultures with drugs, etc.
- DNA: identifies subjects of interest
- RNA: describes activities

Experimental Workflow

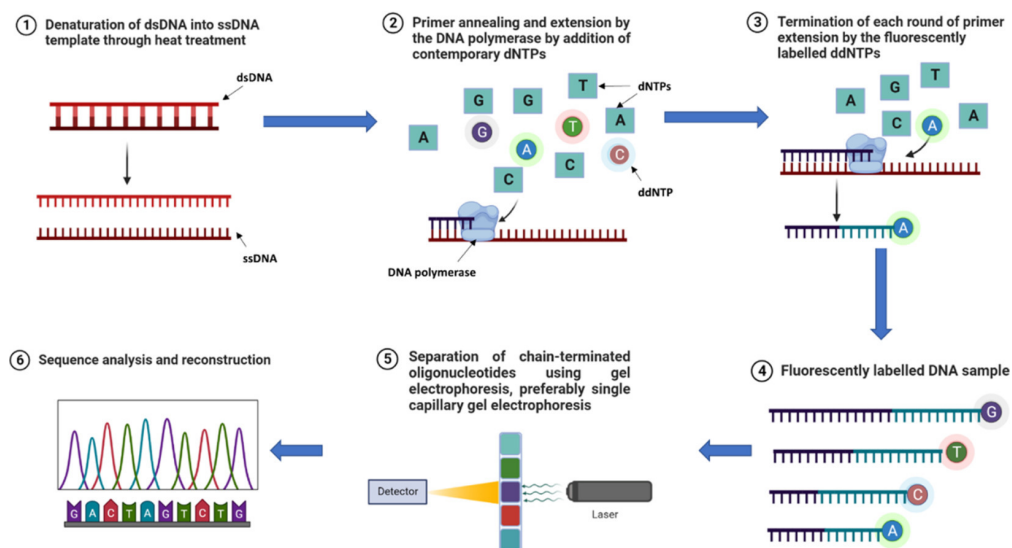


- as data scientists, we start right after sequencing
- we still need to know what was done during samples collection and sequencing
 - “metadata”

Course Goals

- Biological Data Science workflow from seq to data to hypothesis testing
- Interdisciplinary scientific communication
- research projects

DNA sequencing, a molecular approach



- dNTPs, gels (?)
- next generation machines, sequencing by synthesis, fluorophores

Sequence interpretation

- fasta, nt sequence
- RBS
- codons, ORF, start/stop
- we can know genes are there without knowing what they do...

Sequence interpretation by alignment

- reference sequences, databases
- BLASTP
- BWA align
- homology
- function prediction, alphafold

Needleman-Wunsch Alg

(Global Linear Kernel)

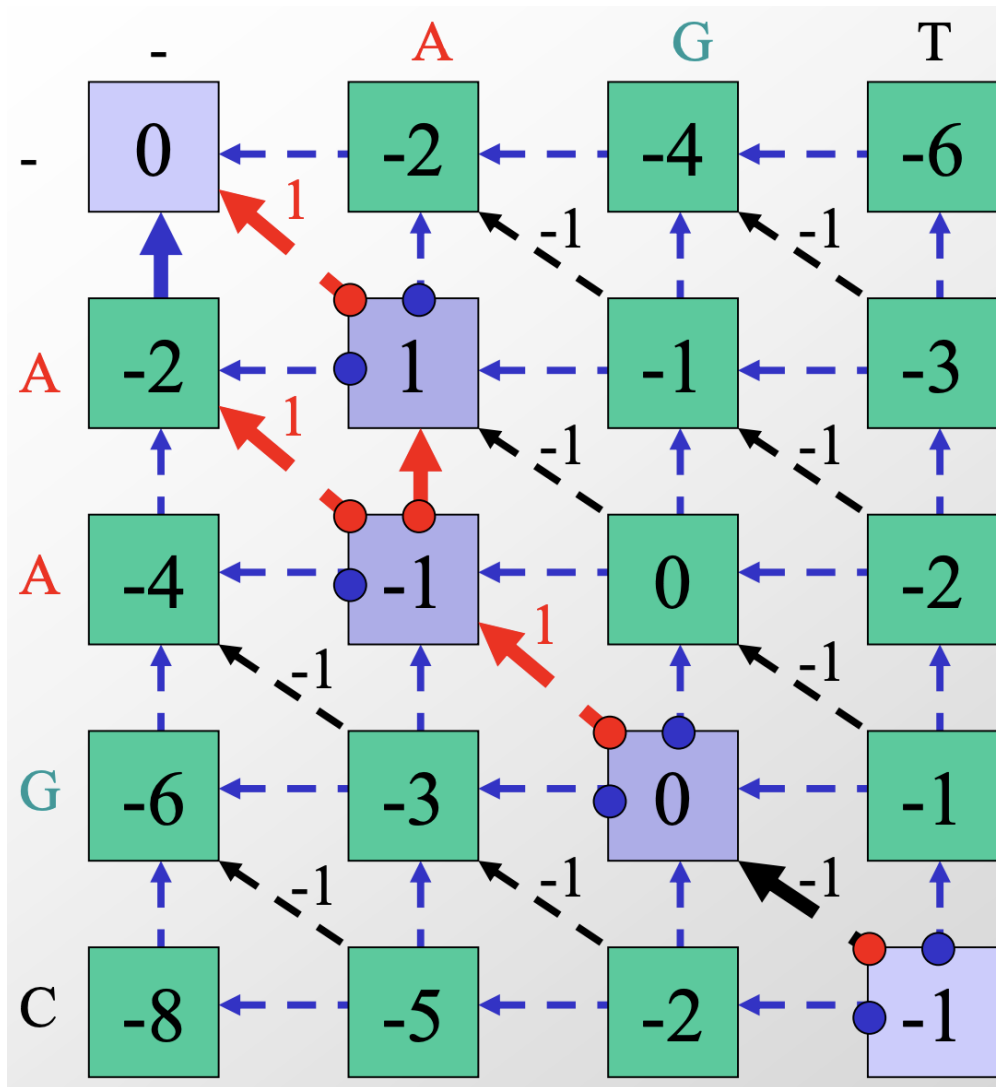
$$S(i, j) = \max \begin{cases} S(i-1, j-1) + s(x_i, y_j) & \text{(Match or Mismatch)} \\ S(i-1, j) + g & \text{(Gap in Y)} \\ S(i, j-1) + g & \text{(Gap in X)} \end{cases}$$

for example, where

$$g = -1match = 1mismatch = -2$$

		G	A	T	T
	0	-2	-4	-6	-8
G	-2	1	0	-2	-4
A	-4	0	2	2	0
T	-6	-2	2	3	4
T	-8	-4	0	4	4

Demo filled



Intructions for the Needleman-Wunsch Alg

Given a mismatch scoring scheme $s(x_i, y_j)$ where x_i and y_i are nucleotides and g is a gap penalty, the optimal alignmnet score of two nucleotide sequences Seq1 and Seq2, indexed by j and i respectively, is :

$$S(i, j) = \max \begin{cases} S(i-1, j-1) + s(x_i, y_j) & \text{(Match or Mismatch)} \\ S(i-1, j) + g & \text{(Gap in Y)} \\ S(i, j-1) + g & \text{(Gap in X)} \end{cases}$$

for example, setting

$$s(x_i, y_j) = \begin{cases} 1 & \text{if Match} \\ -1 & \text{if Mismatch} \end{cases}$$

$$g = -2$$

Then, fill each square as:

$$S(i, j) = \max \begin{cases} \begin{cases} S(i-1, j-1) + 1 & \text{(Match)} \\ S(i-1, j-1) - 1 & \text{(Mismatch)} \end{cases} \\ S(i-1, j) - 2 & \text{(Gap in Y)} \\ S(i, j-1) - 2 & \text{(Gap in X)} \end{cases}$$

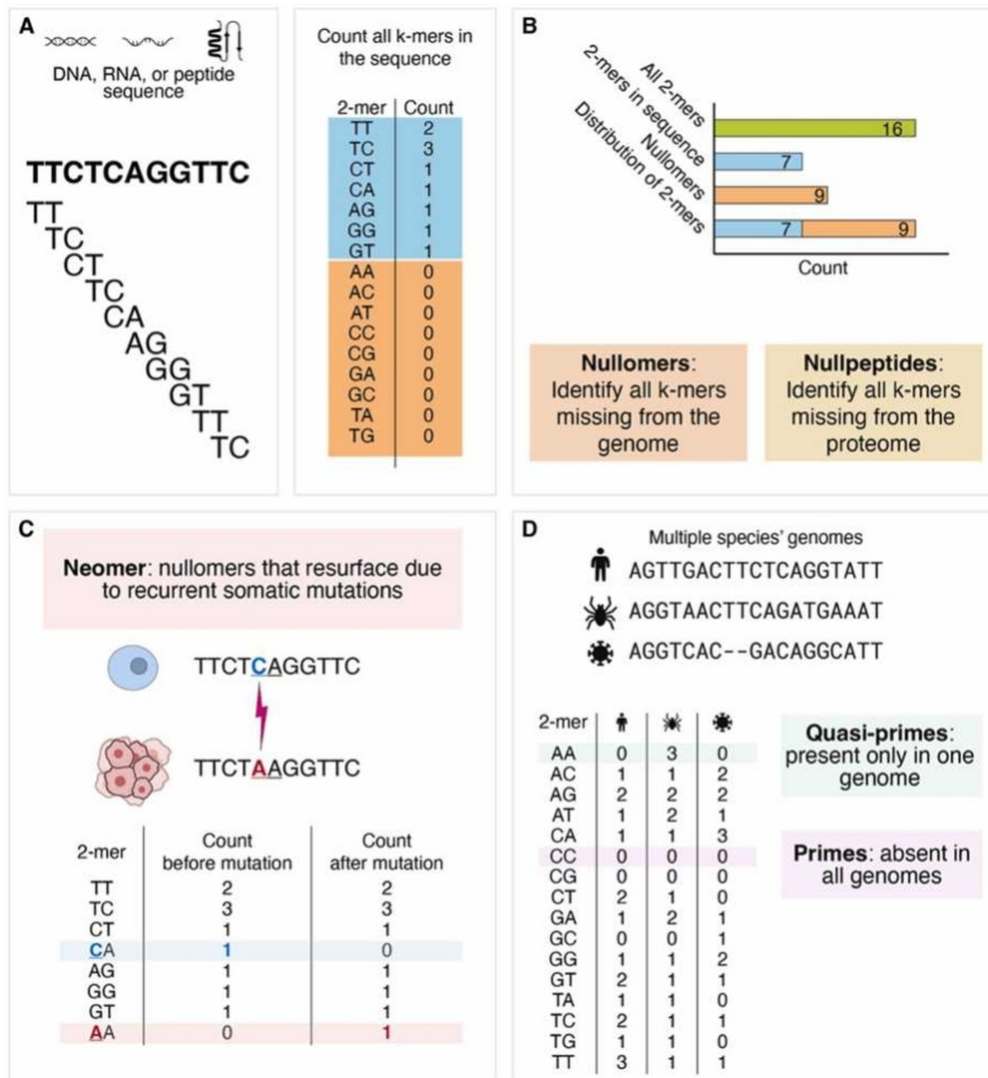
1. Starting in the upper left corner, initiate with 0, then, moving across columns and down rows, fill in each square with:

- the maximum value given the values at the previous indices
- the previous indices from which the value builds (diagonal, left or up arrows)

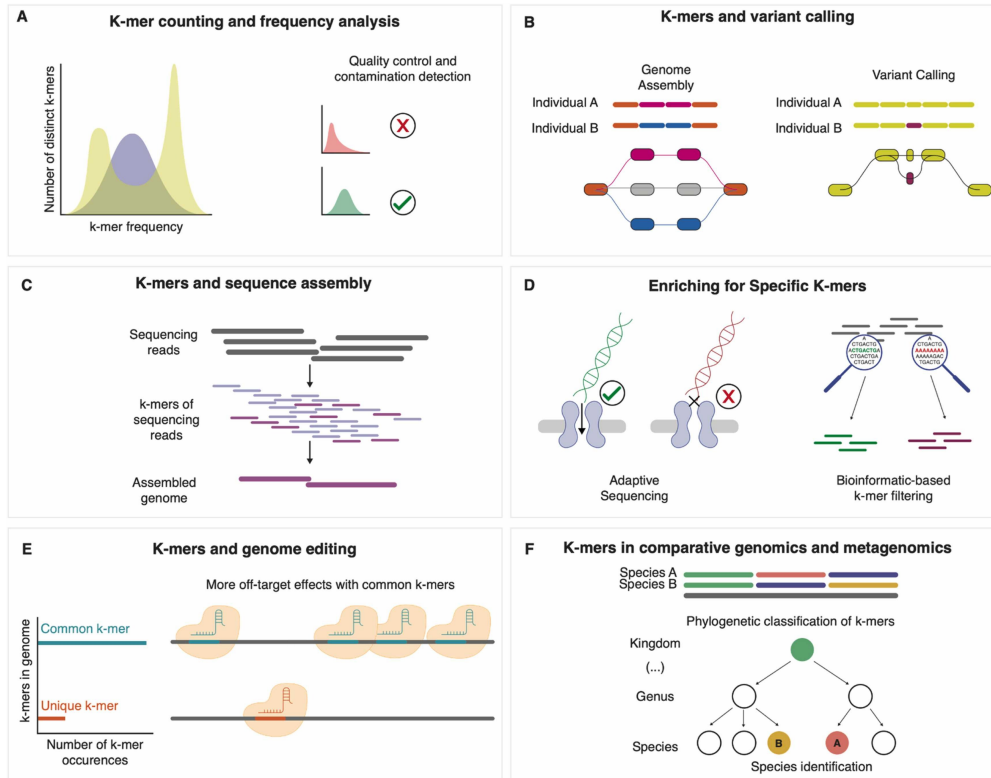
			A	G	T
		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0			
A	$j = 1$				
A	$j = 2$				
G	$j = 3$				
C	$j = 4$				final score

2. Once all squares are filled, **the bottom right corner has your best possible score.**

K-mer based sequence representation



K-mer based quantification



RNA sequencing

- cDNA
- requires reference genomes
- expression levels
- count matrices
- does NOT require annotations

Analysis workflows/goals

- we want to know what cells/organisms use which genes for
- what are gene functions?
- when/where are genes expressed? why?
- are there cause/effect relationships we can interfere with or engineer?

Project goals

- given dataset of E coli gene expression
- what questions can be asked/answered?
- how can metadata be explored? summarized?

Lab machines: getting started

- Log on:
 - Your Union usernames
 - Default password: the word union followed by your ID number; e.g., union12345

Always log off before leaving the lab

Setting up folders & google drive

- Create a folder for this course (e.g. names csc106).
- share it with me (doingg@union.edu)
- request a pro colab account using union email
- find e coli notebook

Rosalind problems

- genes as strings
- accounts
- colab notebooks
- pair up?

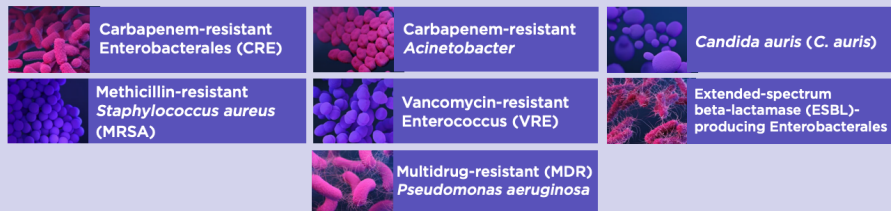
Ecoli metadata

- create colab notebook
- load E coli metadata
- plot a data characteristic accumulation over time
- how does it relate to 'world in data' graph?

E coli background

ANTIMICROBIAL RESISTANCE THREATS in the United States, 2021-2022

CDC used new data¹ to analyze the U.S. burden of the following antimicrobial-resistant pathogens typically found in healthcare settings:



CDC previously reported that the burden of these pathogens increased in the United States in 2020 in the [COVID-19 Impact Report](#). The information below describes the burden in the two following years, 2021 and 2022, and compares against 2019 data.

- strains, genes
- pathogenicity
- CDC <https://www.cdc.gov/ecoli/about/index.html>
 - click to “types”

Activity

- explore metadata
- table
- plot
- revisit: what questions can be asked/answered?

Finish activity: plotting metadata (1 pts)

1. Also in Colab, finish working with the metadata introduced in class.
2. Make at least 3 plots analyzing or summarizing columns.
3. Write down (in Markdown) a comparison you think would be interesting to make, why you think it would be interesting and what you hypothesize the result would look like.
 - you don't have to succeed in your comparison yet, if you'd need more tools/libraries/functions to do so

- it is more important that you have a reason for making the comparison and an estimate to which you could compare your results

HW: rosalind problems

- more python review
- examples of sequences processing/alignment
- at scale, libraries/packages/programs used for these steps
- colab notebook review: practice sharing, commenting, discussing, updating